104.54 EFU COGNITIVE AUTOMATION LICENSE v1.1

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1. DEFINITION OF ALGORITHMS IN THE EFU FRAMEWORK

Within the EFU framework, an **algorithm** is defined as a non-biological decision system that converts information into a course of action or a change in system state.

Cognitive automation therefore does not primarily generate physical labor but **decision flux**.

Fundamental transformation:

Information → Decision → System State

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2. COGNITIVE FLUX BALANCE

The net EFU value of AI-driven cognitive automation is determined by the balance between decision improvement and cognitive entropy.

Key components:

ΔO_decision
Increase in decision quality, measured as improved allocation accuracy and systemic coordination.

S_cognitive
Entropy generated by AI systems, including informational noise, manipulation, dependency creation, or decision distortions.

The EFU balance of cognitive automation therefore depends on whether algorithmic decision systems increase **system-level order** or introduce additional informational entropy.

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3. CRITICAL RISK PATTERNS

3.1 Cognitive Parasitism

An algorithm becomes parasitic when it **extracts human attention without increasing systemic order**.

Examples include:

• dopamine-loop platform architectures
• algorithmic propaganda systems
• engagement-maximizing attention economies

EFU Status: Negative

3.2 Decision Outsourcing (Cognitive Atrophy)

If human decision competence gradually disappears due to excessive reliance on algorithmic systems, the overall system becomes structurally vulnerable.

Examples include:

• fully autonomous weapon systems
• algorithmic governance without human oversight

Such configurations increase systemic fragility by removing the **human adaptive decision layer**.

3.3 Colonial Data Metabolism

A structural risk emerges when **local knowledge systems are extracted and processed in remote infrastructures**.

Risk pattern:

Local knowledge (input) flows to distant data centers where algorithmic processing occurs, resulting in the **loss of local control over decision processes**.

This creates a form of **data colonialism**, where decision power migrates away from the originating community.

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4. EFU AI AUDIT INDICATORS

Indicator	Name	Threshold / Objective
KHA	Cognitive Efficiency Ratio	$\Delta O_decision / (E_compute + S_information) > 1$

Indicator	Name	Threshold / Objective
DFI	Decision Independence Index	$1 - (T_{outsourced} / T_{total_decision}) > 0.7$
EZM	Entropy-Noise Metric	$(S_{noise} + S_{manipulation}) / \Delta O_{decision} < 0.5$
TSI	Transparency Stability Index	Interpretable components / Total model > 0.8
ASK	Data Sovereignty Constraint	Local processing (Edge Computing) > 80%

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5. LICENSING CRITERIA (MINIMUM REQUIREMENTS)

An AI system may receive an **EFU license** only if it satisfies the following **"Golden Triangle" conditions**:

A. Human Competence Preservation (DFI > 0.7)

The system assists human decision-making but does not render human expertise obsolete.

B. Transparency (TSI > 0.8)

The decision logic must remain auditable and interpretable.
Critical system decisions may not rely on **opaque black-box processes**.

C. Local Control (ASK > 80%)

Decision flux and data processing should remain **close to their point of origin** through local or edge computing architectures.

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6. CRITICAL APPLICATION DOMAINS

Healthcare

Diagnostic AI systems capable of significantly reducing medical decision errors and informational entropy.

Ecological Modeling

Simulation systems supporting ecosystem intervention planning and assisting **Module 104.55** in evaluating systemic ecological outcomes.

Disaster Management

AI coordination systems for drone swarms, evacuation optimization, and real-time crisis management to protect **human life flux**.

Status:
Module **104.54 completed and finalized.**